

OOPS PROJECT

ACADEMIA CONNECT

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1. Introduction

The *Academia Connect Student Management System* is a comprehensive software solution designed to manage student-related data efficiently within an academic institution. It integrates multiple modules such as student management, academic tracking, fee management, and reporting into a unified system.

The system ensures secure access, structured data handling, and real-time processing using modern programming concepts like Data Structures and File Handling.

2. Objectives

- To digitize student records and reduce manual work
 - To provide role-based access for students and administrators
 - To manage academic and financial data efficiently
 - To generate performance and analytical reports
 - To ensure data persistence and integrity
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3. System Overview

The system begins with user authentication and directs users to different modules through a **Main Menu**. It supports two main user roles:

- **Student**
- **Admin/Staff**

Each role has controlled access to specific functionalities using role-based access control.

4. System Workflow

4.1 Start and Authentication

- The system starts with credential input.
- Users select their role (Admin, Staff, Student).
- Role-Based Access Control ensures secure entry.

4.2 Student Portal

- Students can access:
 - Profile details
 - Academic grades
 - Fee information

4.3 Backend Data Storage

- All student data is stored in a file (students.dat).
- Data is handled using vectors (e.g., std::vector).
- This ensures structured and efficient data storage.

5. Main Modules

5.1 Student Management

- Performs **CRUD operations**:
 - Create new student records
 - Read existing data
 - Update details
 - Delete records
- Includes:
 - Data validation
 - Use of Data Structures (DSA concepts)

5.2 Academic Tracking

- Manages student grades
- Calculates:

- GPA (Grade Point Average)
 - CGPA (Cumulative GPA)
 - Helps monitor academic performance
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5.3 Fee Management

- Handles financial data:
 - Total fees
 - Paid amount
 - Balance
 - Generates financial statements
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5.4 Reports & Analytics

- Generates performance reports
 - Provides insights into:
 - Academic progress
 - Financial status
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5.5 Settings & System

- Manages:
 - User accounts
 - Backup operations
 - Ensures system reliability and security
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6. Data Persistence Mechanism

- Uses **fstream** for file handling
- Data flow:

- Load data from file → into memory (vector)
 - Modify data → Save back to file
 - Synchronization ensures:
 - No data loss
 - Consistency between memory and storage
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7. Error Handling & Validation

- Input validation prevents incorrect data entry
 - Error handling ensures system stability
 - Covers:
 - Invalid inputs
 - File access issues
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8. System Termination

- Final data is saved before exit
 - File streams are closed
 - Memory is cleared
 - Ensures safe shutdown of the system
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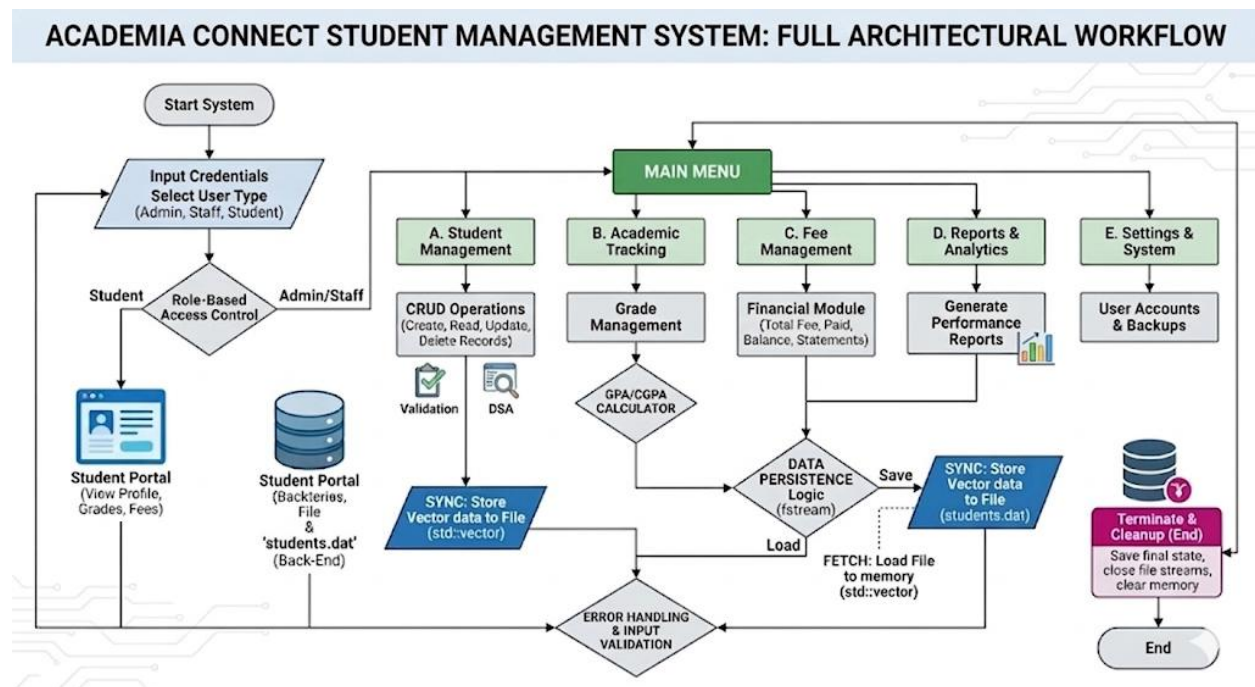
9. Technologies Used

- **Programming Language:** C++
- **Concepts Applied:**
 - Object-Oriented Programming (OOP)
 - Data Structures (Vectors)
 - File Handling (fstream)
 - Modular Design

10. Advantages

- Efficient data management
- Secure role-based access
- Scalable architecture
- Easy report generation
- Reduced manual workload

11. Flow Chart



12. Conclusion

The Academia Connect Student Management System provides a structured and efficient way to manage student information, academic records, and financial data. Its modular architecture, combined with strong data handling and validation mechanisms, makes it a reliable solution for educational institutions.